

STAT 6910 Advanced R

Extra Credit 1

by

Jürgen Symanzik

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UTAH STATE UNIVERSITY

Logan, UT

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1 General Instructions

You can participate in this extra credit (EC) homework in two different roles, depending on your performance in Quiz 1:

Solver: If your score was 10 points or less in Quiz 1, you can work on these questions and turn in your answers for grading. You can obtain 8 EC points at most.

Your answers should be prepared in RStudio, using the combination of R, L^AT_EX, and Sweave (or knitr). Similar to our homeworks, you need to submit the `.Rnw` file and the `.pdf` file produced on your side. **Do not include your names in any of these files. Only name the file names according to the rules for our homeworks!**

Grader: If your score was between 11 and 20 points in Quiz 1, you should still work on these questions, but you receive points only for grading the submissions from the Solvers. You can obtain 4 EC points at most.

Please let me know by e-mail (by the due data) whether you want to participate as Grader in this assignment.

Everyone else can participate as an **Advisor**. You should give hints, either in person or when a question is asked in Canvas. You should not provide a full answer! These EC questions are really intended (and needed) for everyone who did not do so well in Quiz 1.

Depending on how many submissions I get from the Solvers, each Grader may receive one or two such submissions. This will be anonymous, i.e., the Graders won't know whom they are grading and the Solvers won't know who were their Graders.

The score for the Solvers will be determined by the Graders as the median of the reported scores obtained from several Graders. Don't expect to obtain full points this way.

I will only check the work of the Graders in the following way: Graders start with the maximum number of 4 points. For each falsely deducted point when grading a submission, the Grader will be deducted 1/2 point. Moreover, if there was no point deduction made by a Grader, but in case I would have made such a point deduction, the Grader also loses 1/2 point. So, Graders are encouraged to grade carefully. The minimum score for a Grader is 0 points. Graders will have one week to finish grading.

2 Question 1 (3 Points)

The Hermite polynomials are defined recursively by

$$H_0(x) = 1, \quad H_1(x) = x, \quad H_{n+1}(x) = xH_n(x) - nH_{n-1}(x), \text{ for } n \geq 1.$$

Write a function called `CalculateHermite` with arguments `n` and `x`, where `n` is a non-negative integer and `x` can be any numeric value. Do some minimal checks that the values provided by the user meet these requirements.

Calculate the first 6 Hermite polynomials (for x ranging from -5 to 5) and plot them on the same coordinate system. Limit the range of the vertical axis to the interval $[-30, 30]$.

Compare your results with those shown at http://en.wikipedia.org/wiki/Hermite_polynomials. Note that two families of Hermite polynomials exist. Which of these is represented by our recursive formula above? Do the necessary math or carefully compare your figure with the figures shown on the wikipedia page.

3 Question 2 (3 Points)

Recall that

$$\binom{n}{k} = \frac{n!}{k! \cdot (n-k)!} = \binom{n}{n-k}.$$

Instead of using factorials, we can use the following fast calculations, as shown for two examples:

$$\binom{5}{2} = \frac{5!}{2! \cdot 3!} = \frac{5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{(1 \cdot 2) \cdot (3 \cdot 2 \cdot 1)} = \frac{5 \cdot 4}{1 \cdot 2} = ((5/1) \cdot 4)/2 = 10$$

and

$$\binom{1000}{999} = \frac{1000!}{999! \cdot 1!} = \frac{1000!}{1! \cdot 999!} = \frac{1000 \cdot 999!}{1 \cdot 999!} = \frac{1000}{1} = 1000/1 = 1000$$

Write a function called `myBinomialCoeff` with arguments `n` and `k` that calculates Binomial coefficients according to the principle above. Use a while-loop and alternate division and multiplication in the correct order! Note that we won't have to calculate large factorials this way, and, moreover, we never leave the domain of integer numbers during our calculations (if we do this in the correct order).

Your function should check whether `n` and `k` both are valid non-negative integers. If not, return NA and issue a warning.

Test your function with these 2 examples above and add at least 5 more test scenarios, two of which should include 0 for `n` and/or `k`. Compare your results with those obtained from the build-in function `choose`.

4 Question 3 (2 Points)

For the `chip04.txt` data set from the *Graphics 2* chapter of the lecture notes, recreate the plot in the upper left panel of the notes (showing all 4 lines, standardized to 1 in 1975). Your version of this figure should resemble the printed version from the lecture notes as closely as possible.